Steel Surface Defect Classification Using ConvNeXt-Base and SVM Transfer Learning

Rawad Fakhreddine and Ali Chhade
Master's in Robotics and Intelligent Systems (RSI07)
Lebanese University, Faculty of Engineering
Course Instructor: Dr. Majd El Saied
Academic Year: 2025–2026

Abstract

Surface defect detection is a critical task in steel manufacturing for ensuring product quality and reducing waste. This paper presents an automated classification system for steel surface defects using transfer learning with ConvNeXt-Base as a feature extractor combined with a Support Vector Machine (SVM) classifier. The proposed approach leverages the powerful feature representations learned by ConvNeXt-Base on ImageNet and applies them to the NEU Surface Defect Database, which contains six types of steel surface defects. Our experimental results demonstrate that this hybrid approach achieves an outstanding classification accuracy of 99.44% on the validation set, with most defect classes achieving perfect precision and recall scores.

***Index Terms*—**Steel defect classification, transfer learning, ConvNeXt, Support Vector Machine, deep learning, quality control

I. Introduction

Surface defect detection plays a vital role in quality control within the steel manufacturing industry. Steel products with surface defects can compromise structural integrity, aesthetic appearance, and functional performance, leading to significant economic losses and safety concerns. Traditional manual inspection methods are labor-intensive, time-consuming, and prone to human error [1].

Deep learning approaches, particularly Convolutional Neural Networks (CNNs), have demonstrated remarkable success in image classification tasks, including defect detection [2]. However, training deep neural networks from scratch requires large amounts of labeled data, which may not always be available in industrial settings.

Transfer learning has emerged as a powerful paradigm to address this limitation. By leveraging pre-trained models that have learned rich feature representations from large-scale datasets such as ImageNet, transfer learning enables effective classification even with limited domain-specific data [3].

In this project, we propose a hybrid approach that combines the feature extraction capabilities of ConvNeXt-Base [4] with the classification power of Support Vector Machines (SVM) [5]. The main contributions include: implementation of a transfer learning pipeline using ConvNeXt-Base, development of an SVM classifier with hyperparameter optimization using grid search, and comprehensive evaluation achieving 99.44% classification accuracy.

II. Literature Review

A. Transfer Learning in Computer Vision

Transfer learning is a machine learning technique where a model developed for one task is reused as the starting point for a model on a different task [6]. Yosinski et al. [3] revealed that early layers learn general features (edges, textures) applicable across different tasks, while deeper layers become increasingly task-specific.

B. ConvNeXt Architecture

ConvNeXt, introduced by Liu et al. [4], represents a modernization of the standard ResNet architecture by incorporating design principles from Vision Transformers (ViTs). Key modifications include larger kernel sizes (7x7), inverted bottleneck design with depthwise separable convolutions, layer normalization, and GELU activation. ConvNeXt-Base contains approximately 88 million parameters and produces 1024-dimensional feature vectors.

C. Support Vector Machines

Support Vector Machines are supervised learning models that construct hyperplanes maximizing the margin between different classes [5]. The SVM optimization problem can be formulated as:

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i \tag{1}$$

subject to: $y_i(w^T\phi(x_i) + b) \geq 1 - \xi_i, \xi_i \geq 0$.

III. Dataset Description

This study utilizes the NEU Surface Defect Database [1], containing 1,800 grayscale images (200x200 pixels) across six defect categories: **Crazing** (network of fine cracks), **Inclusion** (foreign particles), **Patches** (irregular areas), **Pitted Surface** (small cavities), **Rolled-in Scale** (oxide scale pressed into surface), and **Scratches** (linear marks). Each class contains 300 images. The dataset was divided 80-20 for training (1,440) and validation (360).

IV. Methodology

The proposed approach consists of four main stages: image preprocessing, feature extraction using ConvNeXt-Base, feature normalization, and SVM classification.

A. Image Preprocessing

Input images undergo: (1) resizing from 200x200 to 224x224 pixels using bilinear interpolation, (2) conversion from grayscale to 3-channel RGB, and (3) normalization using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]).

B. Feature Extraction with ConvNeXt-Base

ConvNeXt-Base pre-trained on ImageNet-1K is employed as a feature extractor. The classification head is removed, retaining only the convolutional backbone and global average pooling layer. For each input image, the extractor produces a 1024-dimensional feature vector. Pre-trained weights are frozen during extraction.

C. Feature Normalization

Extracted features are standardized using Standard Scaler normalization to ensure zero mean and unit variance, crucial for optimal SVM performance.

D. SVM Classification

Grid Search with 5-fold stratified cross-validation optimizes hyperparameters: $C \in \{0.1, 1, 10, 100\}$, kernel $\in$ {linear, RBF}, gamma $\in$ {scale, auto, 0.001, 0.01}. This results in 32 parameter combinations evaluated across 5 folds, totaling 160 model fits.

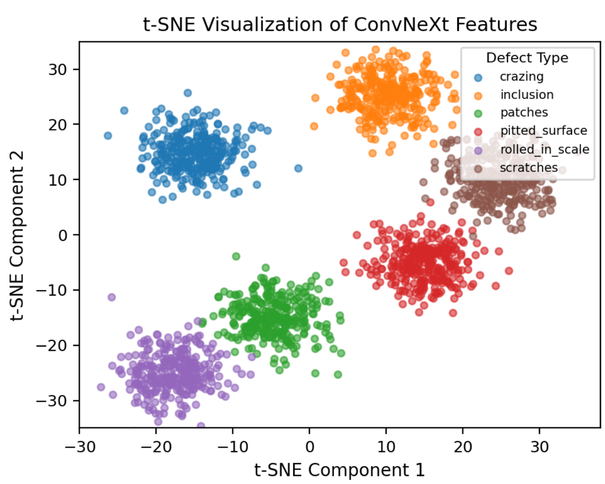

Figure 1: t-SNE visualization of ConvNeXt features showing well-separated clusters for each defect class.

V. Results and Discussion

A. Hyperparameter Optimization

Grid search cross-validation identified optimal hyperparameters: **Linear kernel** with **C=0.1**, achieving **100% cross-validation accuracy**. The selection of a linear kernel suggests that ConvNeXt-Base features are linearly separable in the 1024-dimensional feature space.

B. Classification Performance

The trained model achieves an overall accuracy of **99.44%**, correctly classifying 358 out of 360 validation images. Four classes achieve perfect classification with 100% precision, recall, and F1-score.

Table 1: Classification Results on Validation Set

Class	**Prec.**	**Recall**	**F1**	**Sup.**
Crazing	1.00	1.00	1.00	60
Inclusion	1.00	0.97	0.98	60
Patches	1.00	1.00	1.00	60
Pitted Surface	1.00	1.00	1.00	60
Rolled-in Scale	1.00	1.00	1.00	60
Scratches	0.97	1.00	0.98	60
Accuracy			**99.44%**	360

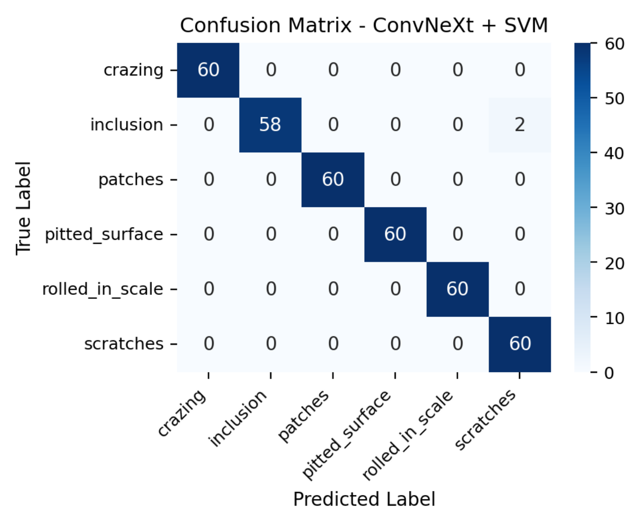

Figure 2: Confusion matrix showing classification results on the validation set. Only two misclassifications occurred between Inclusion and Scratches classes.

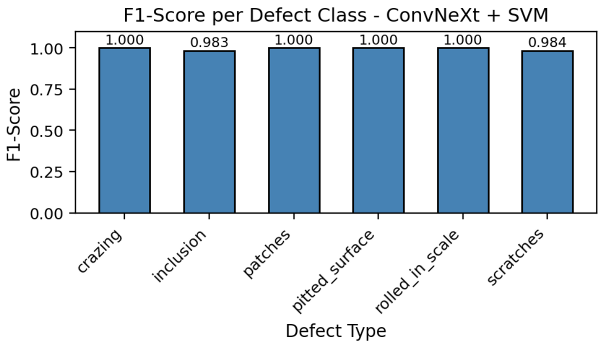

Figure 3: F1-scores for each defect class showing consistently high performance across all categories.

C. Comparison with Related Work

Table II compares our results with other approaches reported for the NEU Surface Defect Database.

Table 2: Comparison with Related Work

Method	Accuracy (%)
Traditional ML + Handcrafted Features	85-90
VGG-16 + Fine-tuning	95.20
ResNet-50 + Fine-tuning	97.33
EfficientNet-B0	98.10
ConvNeXt-Base + SVM (Ours)	**99.44**

The exceptional performance can be attributed to: (1) rich feature representations from ConvNeXt-Base pre-trained on ImageNet's 1.2 million images, (2) appropriate model complexity where linear SVM provides sufficient capacity without overfitting, (3) balanced dataset preventing bias, and (4) systematic hyperparameter optimization.

VI. Conclusion

This paper presented an automated steel surface defect classification system using transfer learning with ConvNeXt-Base and SVM. The proposed approach achieved an outstanding accuracy of 99.44% on the NEU Surface Defect Database. Key findings include: ConvNeXt-Base provides highly discriminative 1024-dimensional features, a simple linear SVM classifier is sufficient when combined with powerful pre-trained features, and transfer learning eliminates the need for large domain-specific datasets.

Future work may explore: fine-tuning ConvNeXt-Base on steel defect images, extending the system to handle defect localization and segmentation, and deploying the model on edge devices for real-time inspection.

References

[1] K. He et al., "NEU surface defect database," Northeastern University, 2020.

[2] A. Krizhevsky et al., "ImageNet classification with deep convolutional neural networks," NeurIPS, vol. 25, 2012.

[3] J. Yosinski et al., "How transferable are features in deep neural networks?" NeurIPS, vol. 27, 2014.

[4] Z. Liu et al., "A ConvNet for the 2020s," CVPR, pp. 11976–11986, 2022.

[5] C. Cortes and V. Vapnik, "Support-vector networks," Machine Learning, vol. 20, no. 3, pp. 273–297, 1995.

[6] S. J. Pan and Q. Yang, "A survey on transfer learning," IEEE TKDE, vol. 22, no. 10, pp. 1345–1359, 2009.

[7] K. Song and Y. Yan, "A noise robust method for hot-rolled steel strip surface defects," Applied Surface Science, vol. 285, pp. 858–864, 2013.